

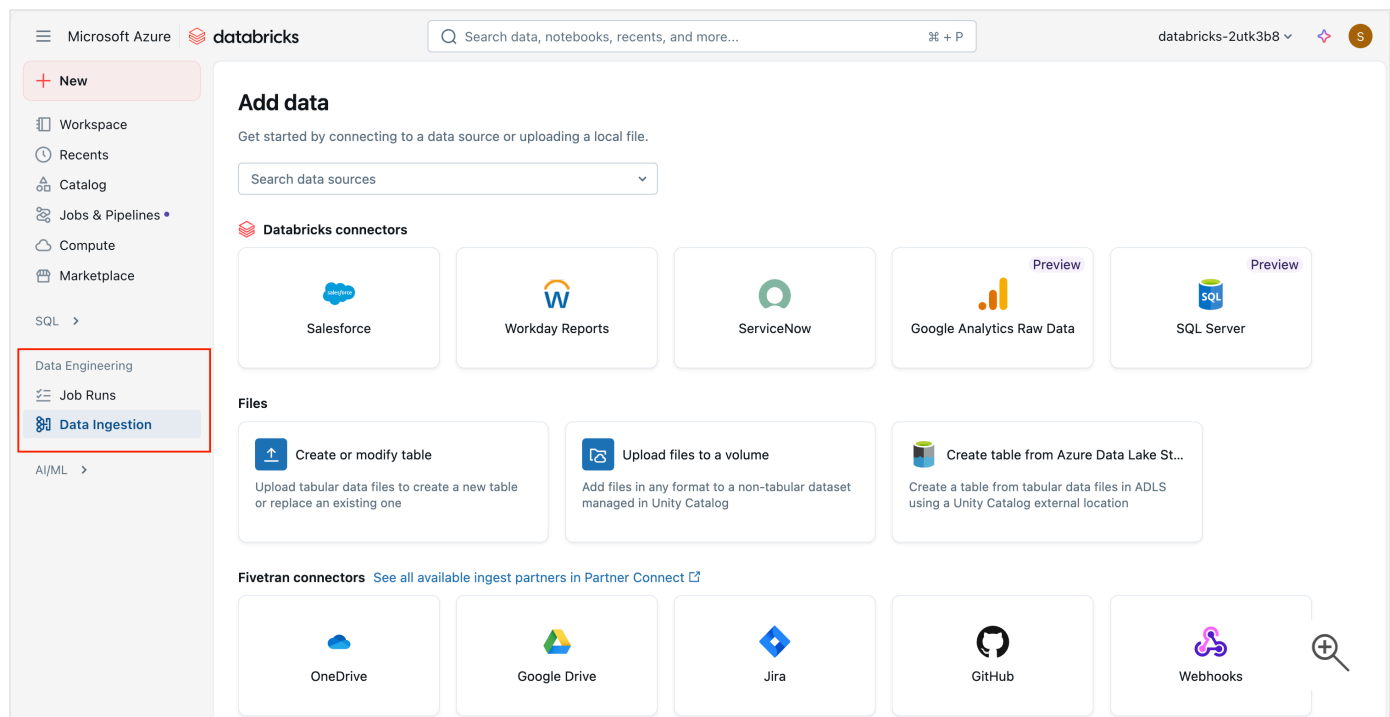
Identify Azure Databricks workloads

3 minutes

Azure Databricks offers capabilities for various workloads including Machine Learning and Large Language Models (LLM), Data Science, Data Engineering, BI and Data Warehousing, and Streaming Processing.

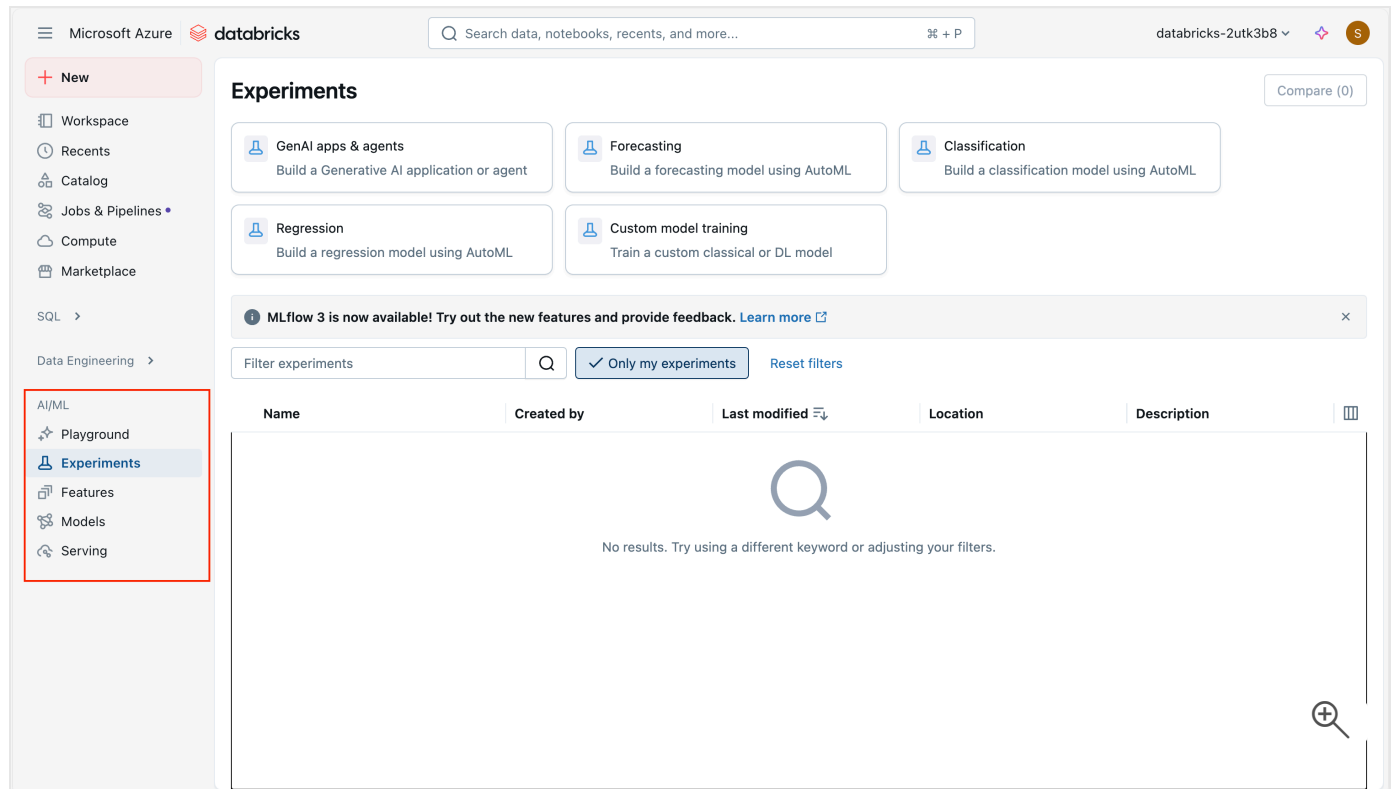
Data Engineering

Azure Databricks provides capabilities for data scientists and engineers who need to collaborate on complex data processing tasks. It provides an integrated environment with Apache Spark for big data processing in a data lakehouse, and supports multiple languages including Python, R, Scala, and SQL. The platform facilitates data exploration, visualization, and the development of data pipelines.



Machine Learning

Azure Databricks supports building, training, and deploying machine learning models at scale. It includes MLflow, an open-source platform to manage the ML lifecycle, including experimentation, reproducibility, and deployment. It also supports various ML frameworks such as TensorFlow, PyTorch, and Scikit-learn, making it versatile for different ML tasks.



SQL

Data analysts who primarily interact with data through SQL can use SQL warehouses in Azure Databricks. The Azure Databricks Workspace UI provides a familiar SQL editor, dashboards, and automatic visualization tools to analyze and visualize data directly within Azure Databricks. This workload is ideal for running quick ad-hoc queries and creating reports from large datasets.

The screenshot shows the Microsoft Azure Databricks interface. On the left sidebar, the 'SQL Editor' option is highlighted with a red box. The main area displays a 'New Query' editor. The top of the interface shows the 'Microsoft Azure' logo, the 'databricks' logo, a search bar, and the user profile 'databricks-2utk3b8'. The 'Catalog' section on the left shows a tree view with 'My organization', 'system', 'databricks_2utk3b8', 'Delta Shares Received', 'samples', 'Legacy', and 'hive_metastore'. The query editor shows a single line of SQL: '1 | SELECT or Option+Space or browse'. Below the query editor, there is a 'Raw results' section with a message: 'No results available. Run a query to show the results.'

Note

SQL warehouses require the Premium tier.

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